



Generative AI Essentials on AWS

Student Guide

Version 1.3.3

100-MLGAIE-13-EN-SG

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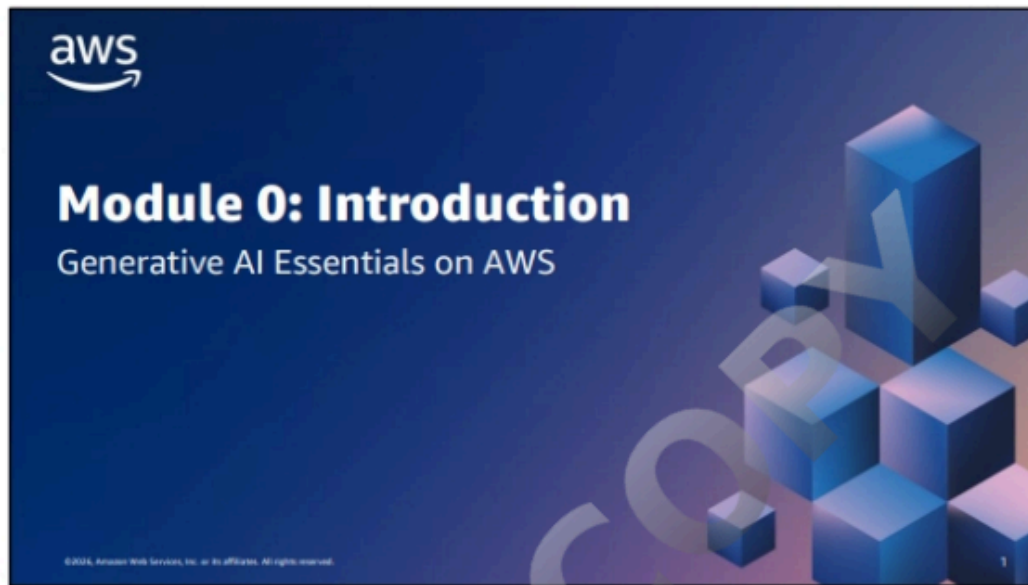
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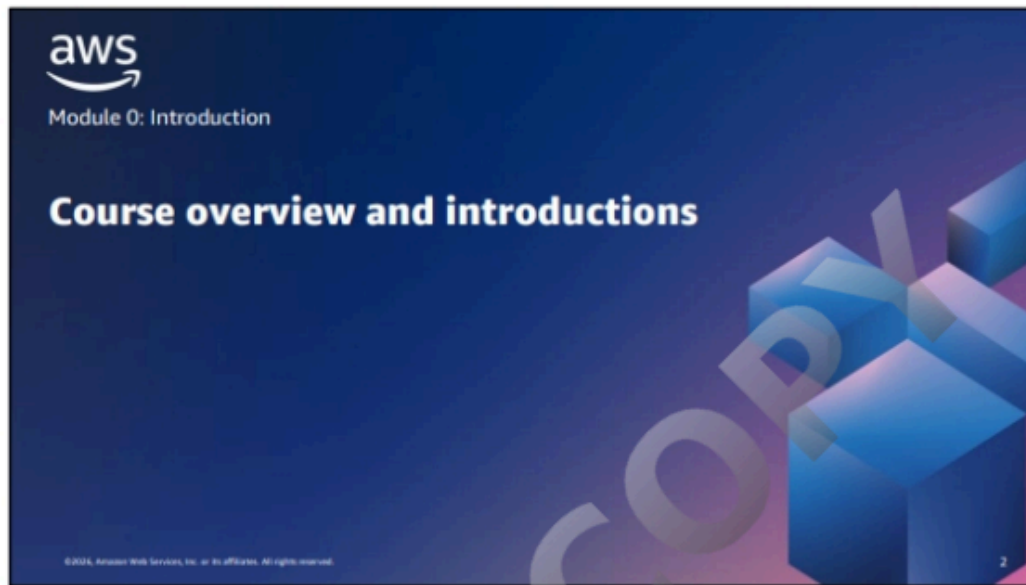
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Course objectives



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On completion, you will be able to do the following:

- Summarize generative AI concepts, methods, and strategies.
- Discuss the appropriate use of generative AI and machine learning and their technologies.
- Describe how to use generative AI responsibly and safely.
- Recognize the types of generative AI solutions with specific use cases.
- Explain implementation and project planning of generative AI to your organization

Agenda

- Module 1: Introducing Generative AI
- Module 2: Exploring Generative AI Use Cases
- Module 3: Essentials of Prompt Engineering
 - Hands-on Lab: Optimizing Slogan Generation with Amazon Bedrock
- Module 4: Responsible AI Principles and Considerations
 - Hands-on Lab: Implementing Responsible AI Principles with Amazon Bedrock Guardrails
- Module 5: Security, Governance, and Compliance
- Module 6: Implementing Generative AI Projects
- Module 7: Integrating Generative AI into the Development Lifecycle
 - Hands-on Lab: Capstone: Creating a Project Plan with Generative AI
- Module 8: Course Wrap-up

Course logistics

- Breaks and lunch
- Security
- Cell phones
- Virtual classroom features
 - Audio
 - Chat
 - Raise hand



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Your instructor reviews expectations for breaks and lunch time. If you are attending your training virtually, review the virtual classroom software features with your instructor. They can explain to you how to use these features for class participation.

Course prerequisites

None

Introduce yourself

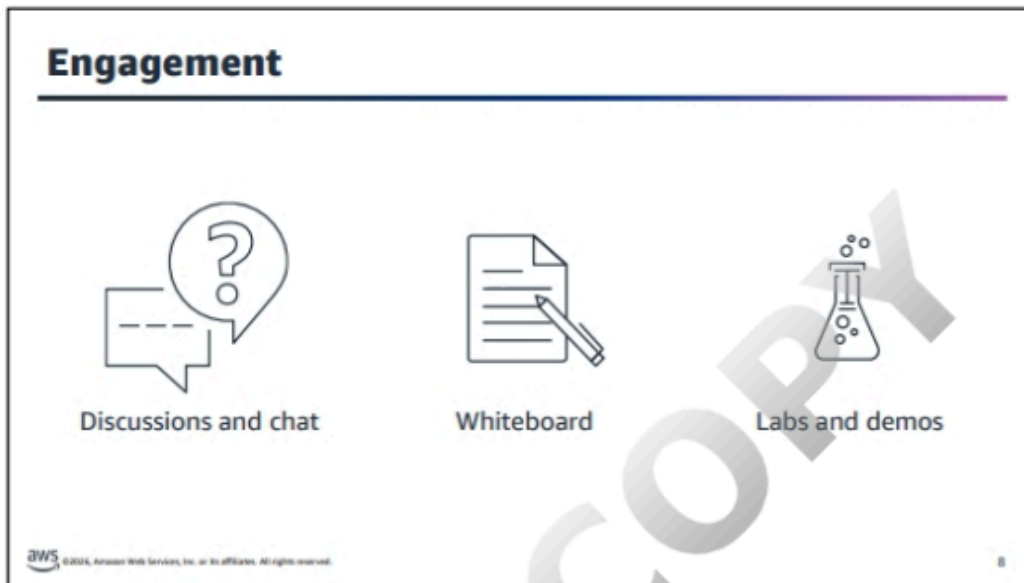
- Name
- What do you do day-to-day?
- What do you want to get out of this class?
- What is your experience level with Generative AI? (1 no experience to 5 expert)
- What is your experience level with AWS?
- Share something personal



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Your introduction helps me understand the experience level of the class and what you want to get out of your time here.



Be an active participant in this course to gain practical experience with generative AI. Use the discussions, chat, whiteboards, and activities to check your knowledge and share your thoughts. Demonstrations will give you additional insight into different aspects of generative AI. Labs will reinforce and apply what you are learning so that you are ready for real-life generative AI.

You are also welcome to ask questions.

Register for access to guides and lab environments

Go to: <https://classrooms.aws.training>

If you **do not have** a SkillBuilder account:

- Create an AWS Builder Labs account by using your Amazon Retail or Partner Central login.

If you **have** a SkillBuilder account: Verify access to lab guides and lab environments.

Create AWS Builder ID

AWS Builder ID is a new personal profile for builders. [Learn more](#)

- ✓ Get started for free
- ✓ Complement your existing AWS accounts
- ✓ Secure your login with optional MFA

Your email address
We recommend using your personal email address.

Next

OR

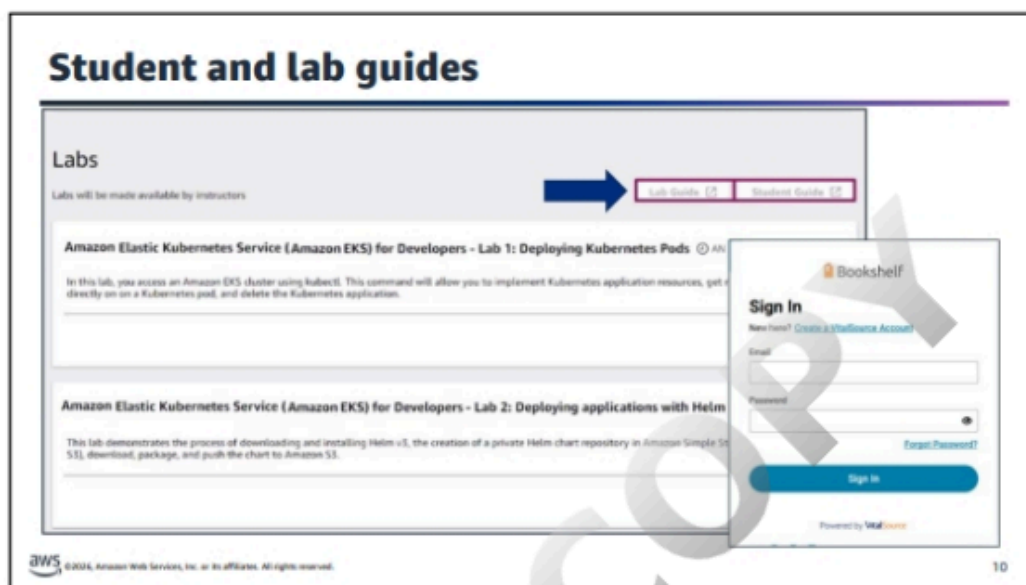
[Already have AWS Builder ID? Sign in](#)

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If you **have** a SkillBuilder account: Verify access to lab guides and lab environments.



At this time, you should be logged in to AWS Builder Labs. From here, you can access your Lab Guide and Student Guide, which are located in eVantage Bookshelf (VitalSource). Buttons to the Lab Guide and Student Guide are located at the upper-right corner of your AWS Builder Labs dashboard. The labs and buttons will be grayed out until the start of the class.

After the class starts, choose either button to access your guides. You will be prompted to log in with your existing eVantage Bookshelf (VitalSource) account or to create a new account. After you are logged in to eVantage Bookshelf (VitalSource), you will have access to the student and lab guides for the class. You can access your guides online or download them. Use these guides to follow along with the course and as a reference after the training.

Lab requirements

Computer running one of the following:

- Windows
- macOS
- Linux: Ubuntu, SUSE, or Red Hat

Recommended web browser:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Reliable internet connection able to browse the internet by using HTTPS

Register for AWS Builder Labs:

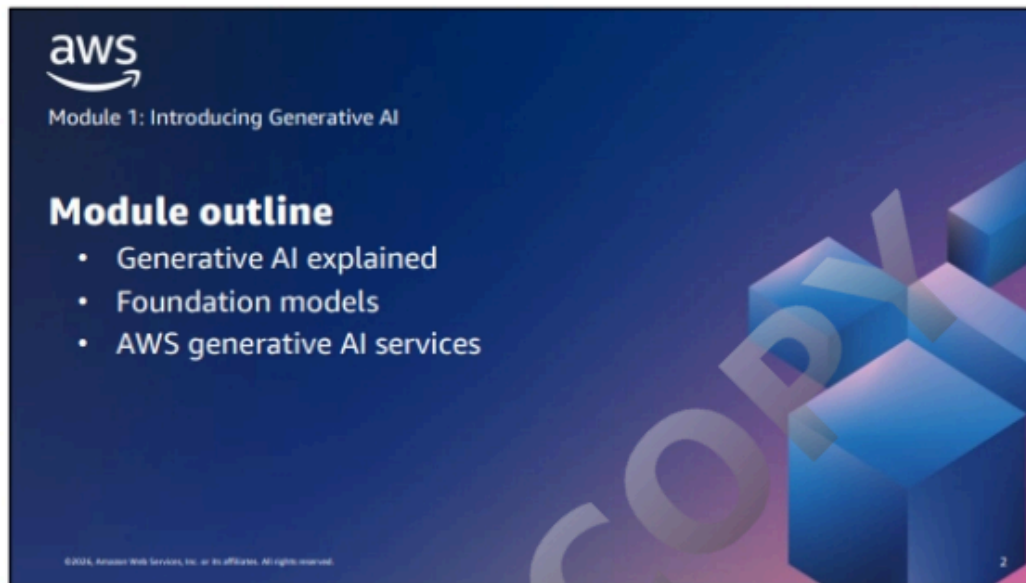
- Turn off ad and script blockers.

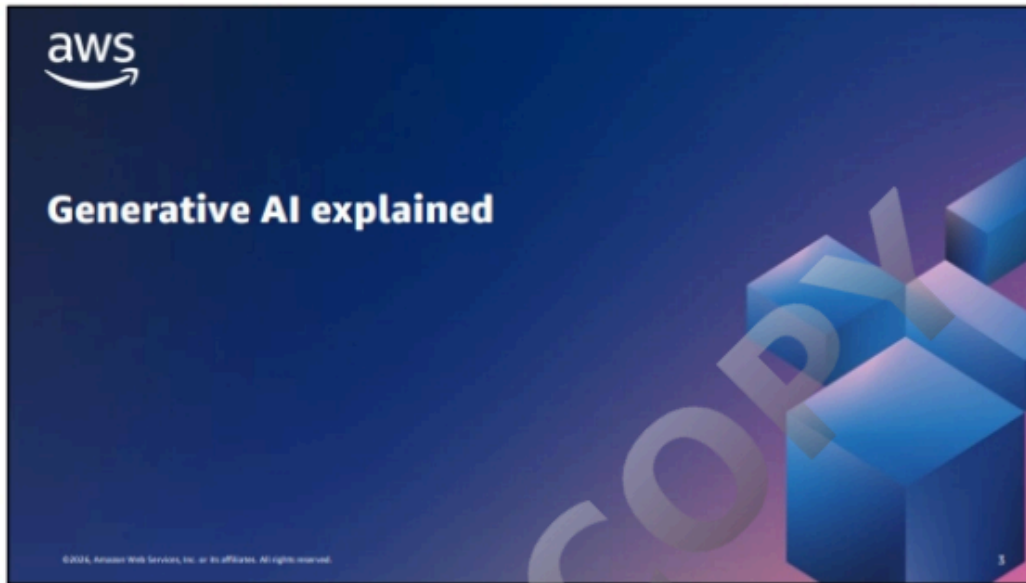


Welcome to the Generative AI Essentials course. Generative AI is a transformative technology. It can create new content across various domains, including text, images, audio, and code. This module introduces core generative AI concepts, capabilities, and enterprise applications.

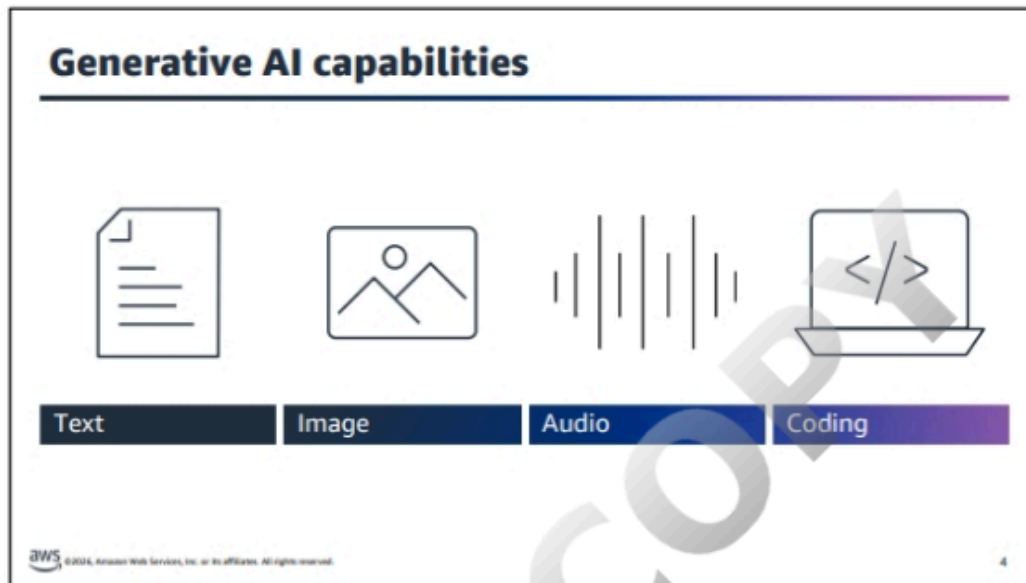
You learn how generative AI enhances creativity, streamlines operations, and improves customer experiences. From automating content creation to powering data analysis, generative AI offers tools to boost productivity and unlock new opportunities. Real-world use cases and success stories illustrate how organizations use this technology for competitive advantages.

Understanding generative AI is crucial for cloud practitioners to stay ahead of technological advancements and provide innovative solutions to clients and stakeholders.





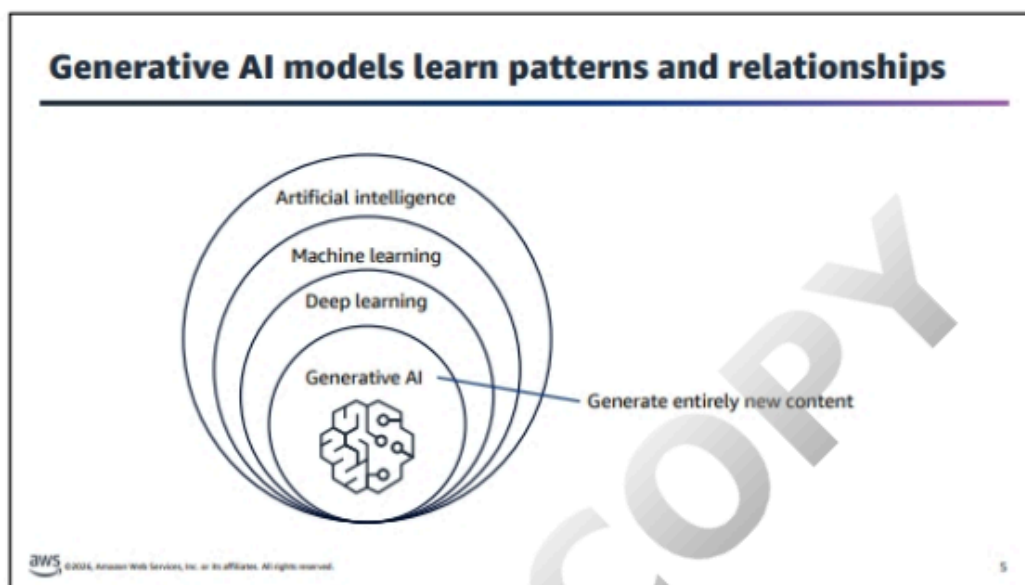
In this section, you explore the concept of generative AI, its relationship to broader AI concepts, and the key differences between generative AI and traditional AI approaches.



Generative AI encompasses a wide range of applications across different content types. In text, generative AI produces human-like writing, from creative stories to technical documentation. For images, it creates, edits, and manipulates visuals, which opens possibilities in design and marketing. In audio, it generates speech, music, and sound effects. Finally, in coding, it assists with writing, debugging, and optimizing software.

These capabilities of generative AI transform industries in several ways. They automate content creation, enhance creative processes, and provide personalized experiences at scale. As you progress in this module, you dive deeper into each area and explore real-world business applications.

Understanding these capabilities is essential for cloud practitioners to identify potential use cases and implement generative AI solutions effectively in cloud environments.

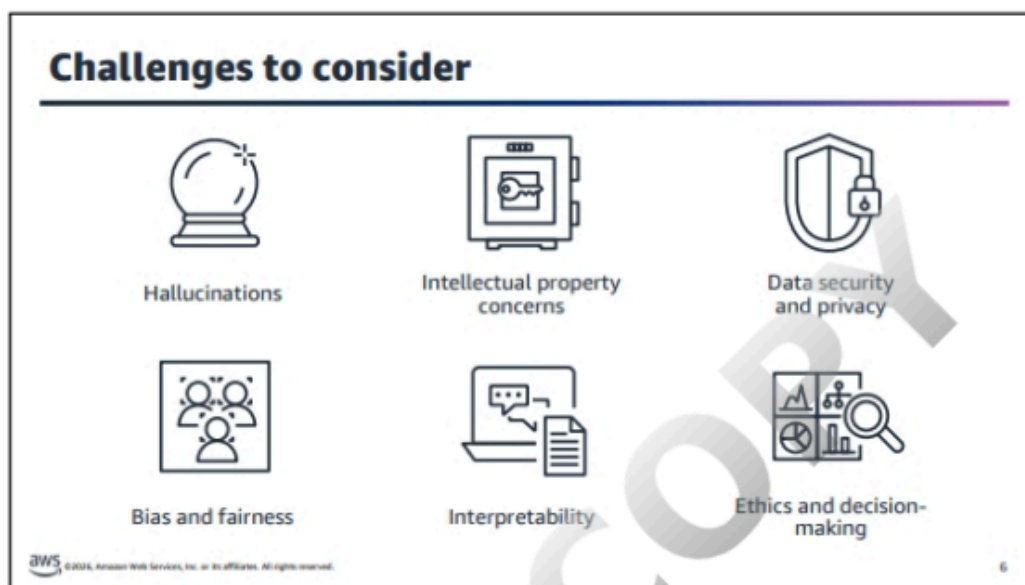


Generative AI models learn patterns and relationships from vast training data, which helps them to create new content that reflects these learned structures. This process differs from traditional AI, which focuses on analysis and prediction.

Understanding generative AI requires knowledge of its relationship to broader AI concepts:

- **Artificial intelligence (AI):** The overarching field of AI, which creates intelligent systems that perform human-like tasks
 - Example: Siri and Alexa are examples of AI systems that can perform human-like tasks such as answering questions, setting reminders, and controlling smart home devices.
- **Machine learning (ML):** A subset of AI that uses statistical techniques for prediction based on patterns
 - Example: Spam filters that learn to identify and block unwanted emails are an example of ML, where the system analyzes patterns in email data to make predictions about future messages.
- **Deep learning (DL):** A type of ML based on neural networks that are capable of learning complex patterns from large datasets
 - Example: Facial recognition systems used in smartphones and social media platforms are powered by deep learning, which can learn complex patterns in large datasets of facial images.
- **Generative AI:** A subset of DL that creates new data based on learned patterns, often without retraining
 - Example: Text-generating models like Amazon Nova Lite and image-generating models like Amazon Nova Canvas are examples of generative AI, which can create new content (such as articles, stories, or images) based on the patterns they've learned from their training data.

This hierarchical relationship shows how generative AI extends predecessor technologies, and offers unprecedented potential for content creation and problem-solving across domains.



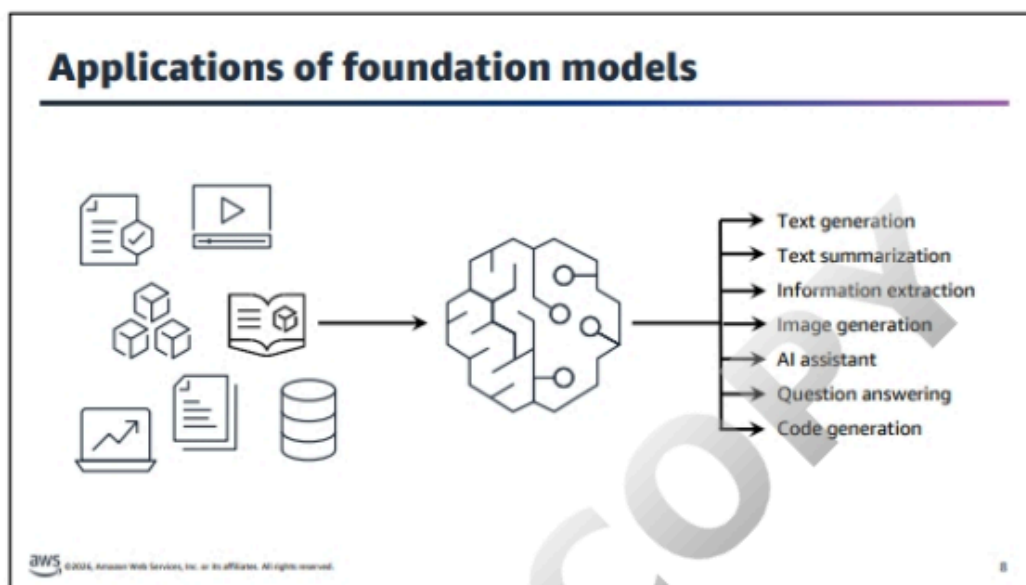
While generative AI offers immense potential, it also presents significant challenges that you must carefully address:

- **Hallucinations:** AI models can sometimes generate false or misleading information, which poses risks to reliability and trustworthiness. To mitigate these risks, you can implement fact-checking mechanisms and maintain human oversight.
- **Intellectual property (IP) concerns:** The use of copyrighted material in training datasets and the potential for AI-generated content to infringe on existing IP rights raise complex legal and ethical questions. To help address these issues, carefully curate training data and use content filtering systems.
- **Data security and privacy:** The vast amounts of data required for AI models create potential vulnerabilities for breaches and privacy violations. Taking robust measures for data protection and adhering to privacy regulations are crucial.
- **Bias and fairness:** AI models can perpetuate or amplify existing societal biases, which can lead to unfair outcomes. To help mitigate these issues, perform regular audits of model outputs, and provide diverse representation in training data.
- **Interpretability:** The lack of transparency in how many AI models operate makes it difficult to understand their decision-making processes. Developing explainable AI techniques and maintaining transparency in model deployment are essential for accountability and trust.
- **Ethics and decision-making:** It is increasingly important to make sure that AI systems make decisions that are aligned with human values and ethical principles. It is crucial that you establish clear ethical guidelines and implement governance frameworks.

To address these challenges, you must take a multifaceted approach involving technical solutions, policy development, and ongoing education. You must prioritize responsible AI practices and maintain human oversight to make sure the technology benefits society while minimizing potential harm.



Foundation models are a crucial component of generative AI because they provide a versatile and powerful foundation for transfer learning. With transfer learning, models can be adapted and fine-tuned for a wide range of tasks and applications.



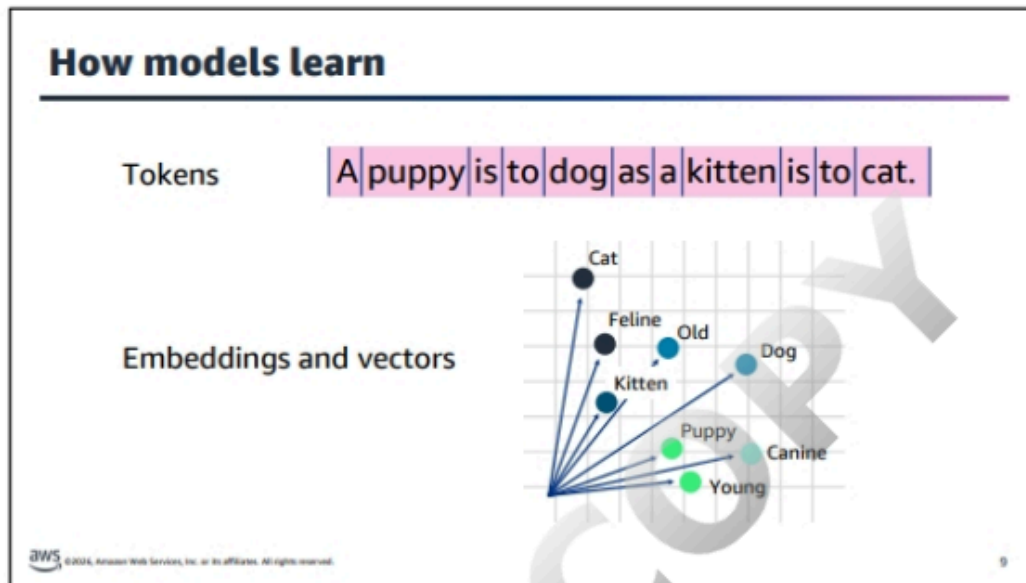
Foundation models demonstrate remarkable versatility and can perform a wide range of tasks, including the following:

- **Text generation:** Creating coherent, contextually appropriate text for various purposes
- **Text summarization:** Condensing long documents into concise summaries
- **Information extraction:** Identifying and extracting specific data points from unstructured text
- **Image generation:** Creating new images based on textual descriptions or inputs
- **AI assistants:** Powering conversational AI systems for customer service and interactions
- **Question answering:** Providing accurate responses based on available information
- **Code generation:** Automatically generating code for various programming languages and tasks based on natural language descriptions or specifications

Adapting foundation models involves two steps:

1. **Pre-training:** Initial training on large, diverse datasets to capture general knowledge
2. **Adaptation:** Fine-tuning or prompting the model for specific tasks or domains

By using this approach, businesses can use powerful AI capabilities without extensive data collection or model development, which significantly accelerates AI adoption and innovation.



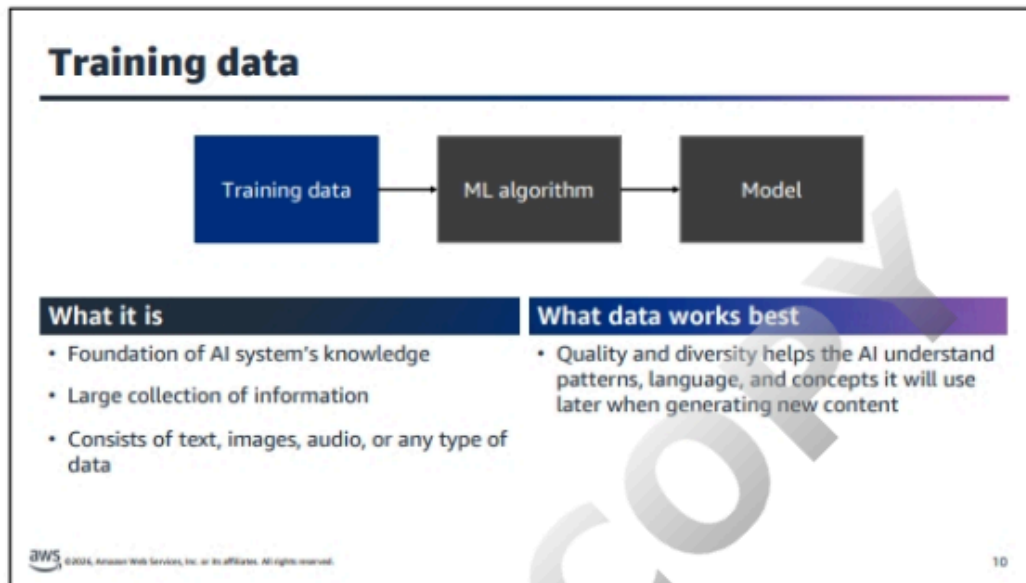
To understand how generative AI processes and creates text, three key concepts are crucial: large language models (LLMs), tokens, and embeddings.

LLMs are advanced AI systems that are trained on vast text data. LLMs use complex neural architectures such as transformers to understand and generate human-like text. They capture intricate language patterns and relationships, and provide diverse text-based capabilities.

Tokens are the basic units that LLMs process—words, word parts, or characters. For example, the sentence "A puppy is to dog as a kitten is to cat" might be tokenized into individual words.

Embeddings numerically represent tokens in high-dimensional spaces, with each token assigned a vector that captures its meaning and relationships. This allows models to understand semantic similarities, for example, "cat" being closer to "kitten" than "dog."

These concepts work together for LLMs to process and generate text effectively by representing language in structured ways that capture complex linguistic patterns.

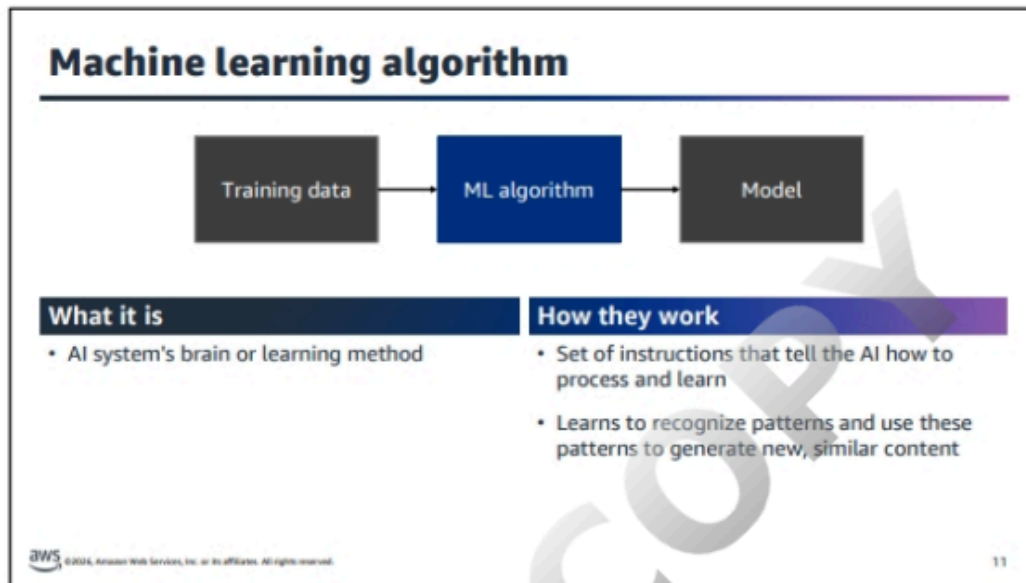


Training data serves as the foundation of an AI system's knowledge. Training data is a large collection of information input into the AI during the training process. This data consists of text, images, audio, or any other type of data.

The quality and diversity of this training data are crucial factors that impact an AI model's performance. An AI system, like a student, performs best when trained on high-quality, diverse datasets. This exposure enhances its learning and performance, and helps the AI understand patterns, language, and concepts that it will use when generating new content.

You can use both structured data (organized in predefined formats) and unstructured data (lacking predefined structures) to train AI models. However, unstructured data often requires more advanced techniques for processing and analysis.

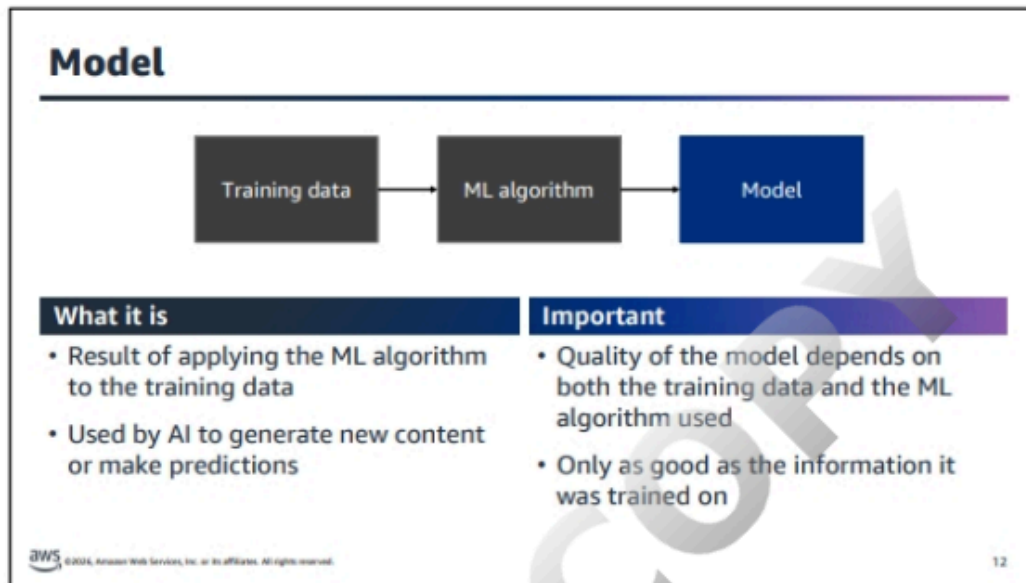
By carefully curating and diversifying training datasets, organizations can develop AI systems with broad knowledge bases. The AI systems will also have the ability to generate relevant, contextually appropriate outputs across a wide range of applications and domains.



Machine learning algorithms are the core components that AI systems use to learn from data and make predictions or generate new content. In the context of generative AI, these algorithms are designed to recognize patterns in the training data and use these patterns to create new, similar content.

There are various types of ML algorithms used in generative AI, with neural networks being particularly prevalent. These algorithms, inspired by the human brain's structure, can learn complex patterns and relationships in data.

The algorithm that you choose greatly affects model performance and capabilities. Consider factors such as the nature of the task, available data, and computational resources when selecting an algorithm. AWS services such as SageMaker AI provide a range of pre-implemented algorithms and the flexibility to use custom algorithms. Therefore, you can select the most suitable approach for your specific generative AI projects.

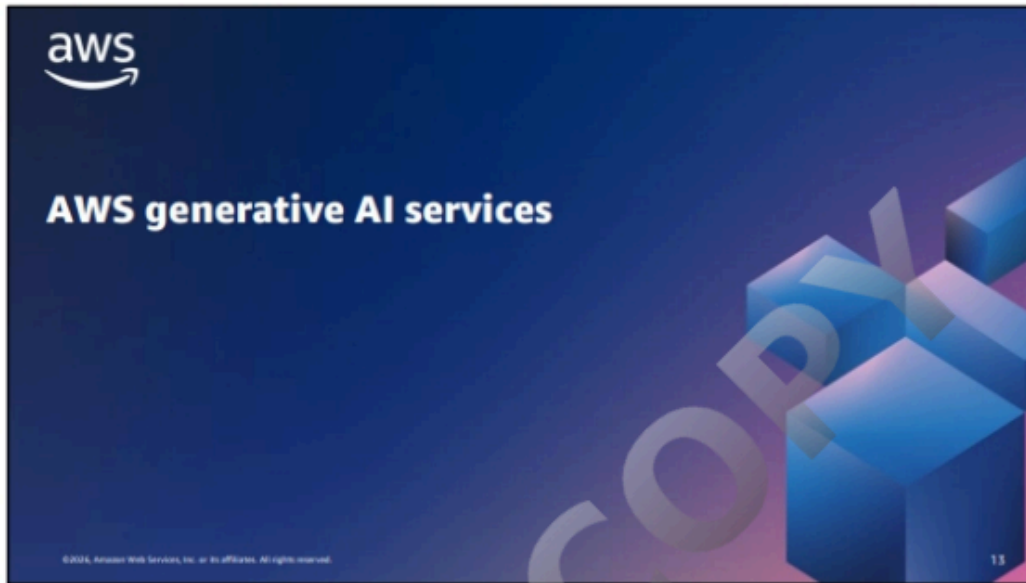


In generative AI, a model is the result of applying a machine learning algorithm to training data. Models encapsulate the patterns, relationships, and rules learned from the data, so that the AI system can generate new content or make predictions when given new inputs.

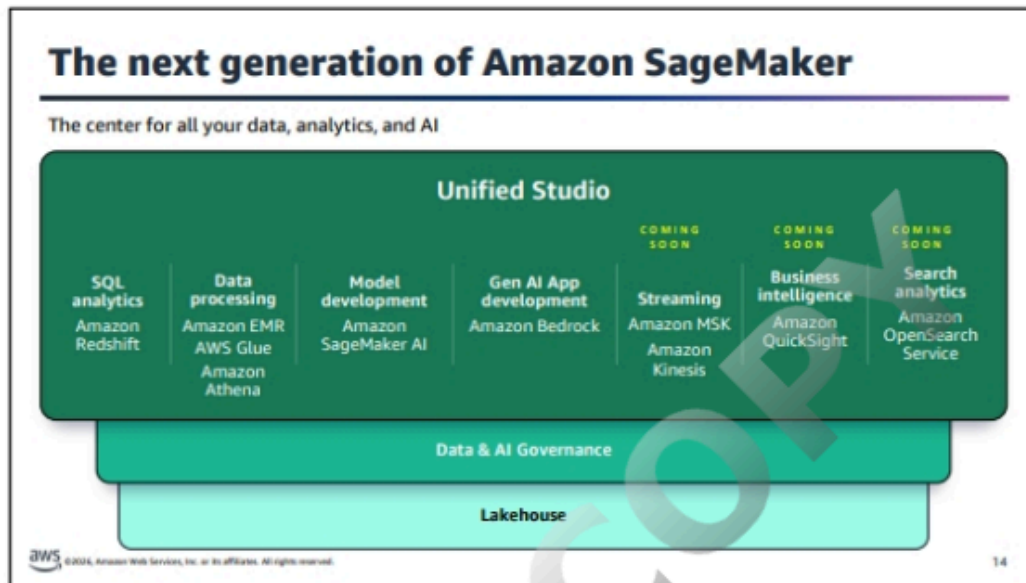
The quality of a generative AI model is critically dependent on both the training data and the ML algorithm that you use. High-quality, diverse training data helps the model learn a wide range of patterns and nuances, while an appropriate algorithm ensures effective learning from this data.

Model development is often iterative. Initial models might have limitations or biases. You can address these issues by refining training data, adjusting algorithms, or fine-tuning model parameters. AWS services such as SageMaker AI help with this iterative process by providing tools for model training, evaluation, and deployment.

Be aware that a model is only as good as the information it was trained on. It is important to carefully curate data and continuously monitor and update models to make sure they remain accurate and relevant over time.



In this section, you learn about the AI/ML service offerings from AWS, including key generative AI services such as Amazon Bedrock and Amazon SageMaker JumpStart.



You may have heard that AWS recently made big changes to Amazon SageMaker. Amazon SageMaker is used by hundreds of thousands of AWS customers to build, train, and deploy machine learning models. Now, we've taken the machine learning service and added AWS analytics capabilities - creating one unified platform for data, analytics, and AI.

The next generation of Amazon SageMaker includes virtually all of the components you need for fast SQL analytics, big data processing, search, data preparation and integration, AI model development and training, and generative AI – along with a single view into all of your enterprise data. You get a single data and AI development environment with the SageMaker unified studio, a lakehouse architecture that unifies access to all your data - on S3, in Redshift, in SaaS applications, on-premises, or in other clouds - through the open Apache Iceberg standard interface and with the SageMaker Catalog built into Unified Studio, you get end-to-end governance for your data and AI workflows.

The next generation of Amazon SageMaker is generally available today via the AWS console, and you can learn more at www.amazon.aws.com/sagemaker.



Amazon SageMaker AI

The service previously known as Amazon SageMaker has been renamed Amazon SageMaker AI. It is integrated within the next generation of SageMaker and is also available as a standalone service for those who wish to focus specifically on building, training, and deploying AI and ML models at scale.

Amazon SageMaker AI is a fully managed service to build, train, and deploy ML models – including foundation models – for any use case by bringing together a broad set of tools to enable high-performance, low-cost machine learning. It is available as a standalone service in the AWS console, or via APIs. Model development capabilities from SageMaker AI are available in the next generation of Amazon SageMaker.

1/Amazon SageMaker AI provides access to high-performance, cost-effective, scalable, and fully managed infrastructure and tools for each step of the ML lifecycle. Using Amazon SageMaker AI tools, you can easily build, train, test, troubleshoot, deploy, and manage FMs models at scale and boost productivity of data scientists and ML engineers while maintaining model performance in production.

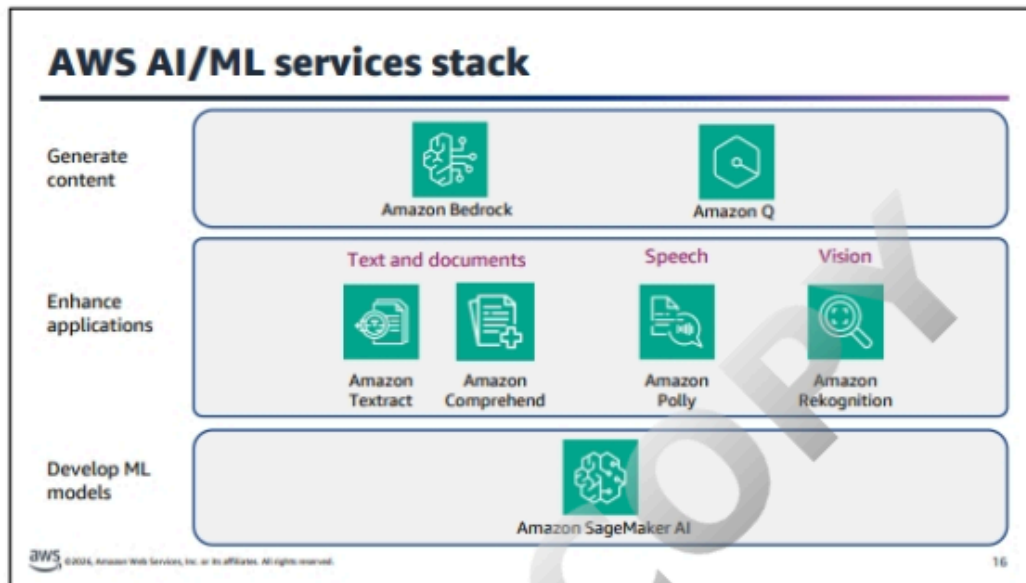
2/You can explore Amazon SageMaker JumpStart, which is a ML hub offering models, algorithms, and prebuilt ML solutions. SageMaker JumpStart offers hundreds of ready-to-use FMs from various model providers, including a growing list of best performing publicly available FMs such as Falcon-40B, Stable Diffusion, OpenLLaMA, and Flan-T5/UL2.

3/Amazon SageMaker machine learning operations (MLOps) capabilities help you create repeatable workflows across the ML lifecycle to experiment, train, deploy, and govern ML models at scale while maintaining model performance in production.

4/Amazon SageMaker AI provides purpose-built governance tools to help you implement ML responsibly. Amazon SageMaker Model Cards makes it easier to capture, retrieve, and share essential model information. Once the models are deployed, SageMaker Model Dashboard gives you unified monitoring across all your models by providing deviations from expected behavior, automated alerts, and troubleshooting to improve model performance. Amazon SageMaker Clarify detects and measures potential bias using a variety of metrics to help you address potential bias and explain model predictions.

5/With Amazon SageMaker Ground Truth, you can use human feedback to customize models on company- or domain-specific data for your unique use case to improve model output and task performance.

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AWS offers a comprehensive set of AI/ML services that provide solutions at various abstraction levels. At the foundation level is ML frameworks such as Amazon SageMaker AI, a fully managed platform for building, training, and deploying machine learning models.

The next layer consists of AI/ML services that address specific use cases, such as natural language processing (Amazon Comprehend), computer vision (Amazon Rekognition), and conversational AI (Amazon Lex). These services provide AI integration without the need to build models from scratch.

At the top level, is generative AI services, including Amazon Bedrock for accessing foundation models, SageMaker JumpStart for prebuilt solutions, and Amazon Q for AI-powered assistance across business contexts.

This layered approach offers flexibility. Businesses can choose to build custom models or quickly integrate prebuilt AI capabilities, depending on their needs. By combining services across layers, organizations create powerful, end-to-end AI/ML solutions using each component's strengths.

For example, a company might use SageMaker AI to train a custom model and integrate Amazon Comprehend for text analysis. They might then use Amazon Bedrock to generate human-like responses based on combined insights. With this flexibility, businesses can tailor AI/ML solutions to specific requirements and use cases.

Sample of Amazon Bedrock pre-trained model options



Amazon
Bedrock

Amazon: Nova

Anthropic: Claude

AI21 Labs: Jamba models

Stability AI: Stable Diffusion

Cohere: Command

Meta: Llama

Mistral AI: Mistral

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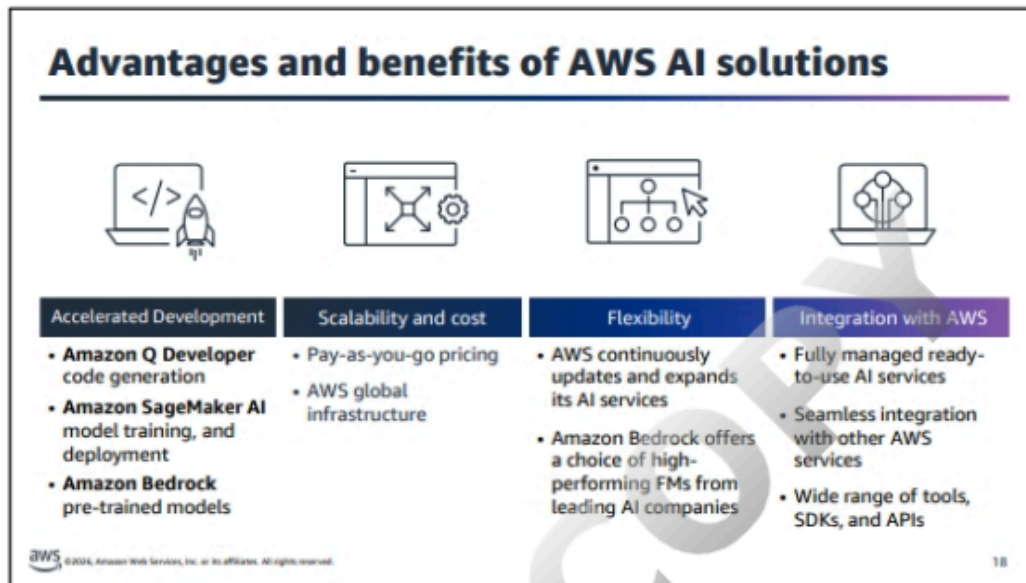
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Amazon Bedrock is a fully managed service that provides access to a wide range of foundation models from leading AI companies. This slide shows a sample of the pre-trained model options available through Amazon Bedrock, which highlights the diversity and flexibility of the platform.

Each model provider has unique strengths. Amazon Nova is a collection of foundation models (FMs) that use generative AI to create images, videos, and text. These models are designed to be fast, cost-effective, and easy to use. Anthropic's Claude models are known for their strong reasoning capabilities and ethical considerations. AI21 Labs' Jamba models excel in long-form text generation and summarization.

Stability AI's Stable Diffusion models specialize in image generation and manipulation, while Cohere's Command models focus on natural language understanding and generation. Meta's Llama 3 models offer state-of-the-art performance across various language tasks, and Mistral AI's models are known for their efficiency and adaptability.

When choosing a model, consider task requirements, performance, cost, and each model's strengths and limitations. These factors will help you find the best fit. With Amazon Bedrock, you can experiment with different models to find the best fit for your needs, all through a single, consistent API. This flexibility is crucial in the rapidly evolving field of generative AI, and you can stay current with the latest advancements without constantly retooling your infrastructure.



AWS AI solutions offer several key advantages for organizations implementing generative AI:

Accelerated Development – Tools such as Amazon Q Developer for code generation and Amazon SageMaker AI for automated ML workflows significantly speed up development cycles.

Scalability and Cost Optimization: Pay-as-you-go pricing and the AWS global infrastructure provide efficient scaling of AI workloads.

Flexibility and Access to Models: Continuous updates to AI services and a wide selection of foundation models through Amazon Bedrock provide cutting-edge capabilities.

Integration with AWS Ecosystem: Seamless integration with other AWS services provides end-to-end AI/ML solutions.

Security and Compliance: Robust security features and compliance certifications help protect sensitive data and meet regulatory requirements.

Because of these advantages, you can innovate faster, reduce costs, and use the latest AI technologies while maintaining security and scalability.

Demonstration

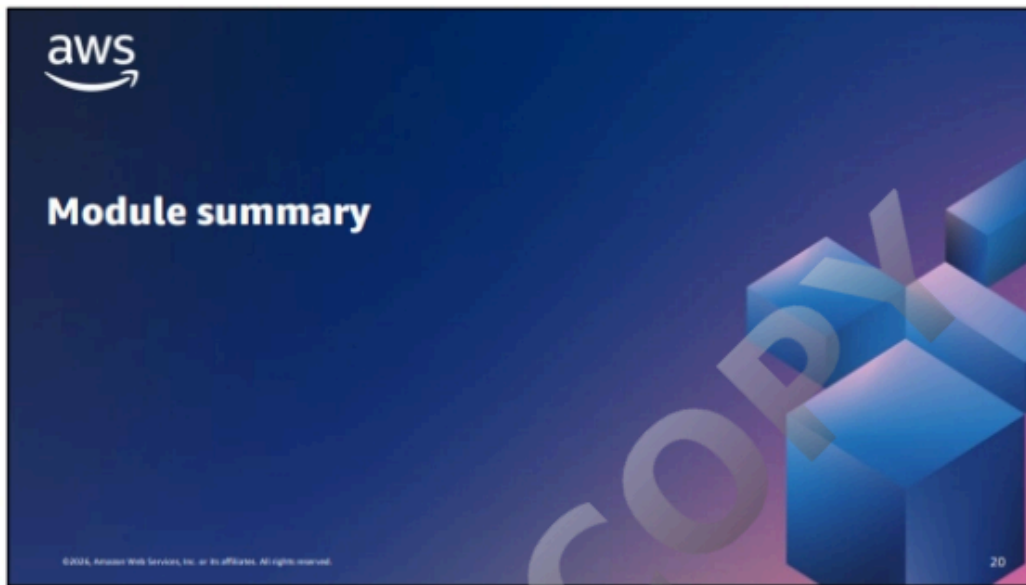


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Generative AI solution

In this demo, your instructor will show you the following things:

- Generate product description.
- Generate images.
- Interact with chat interface.



This section recaps the main points covered in the module and highlights the key takeaways about generative AI and the AWS services that support it. You are encouraged to engage in questions and discussions to deepen your understanding of this exciting and rapidly evolving field.

Module summary



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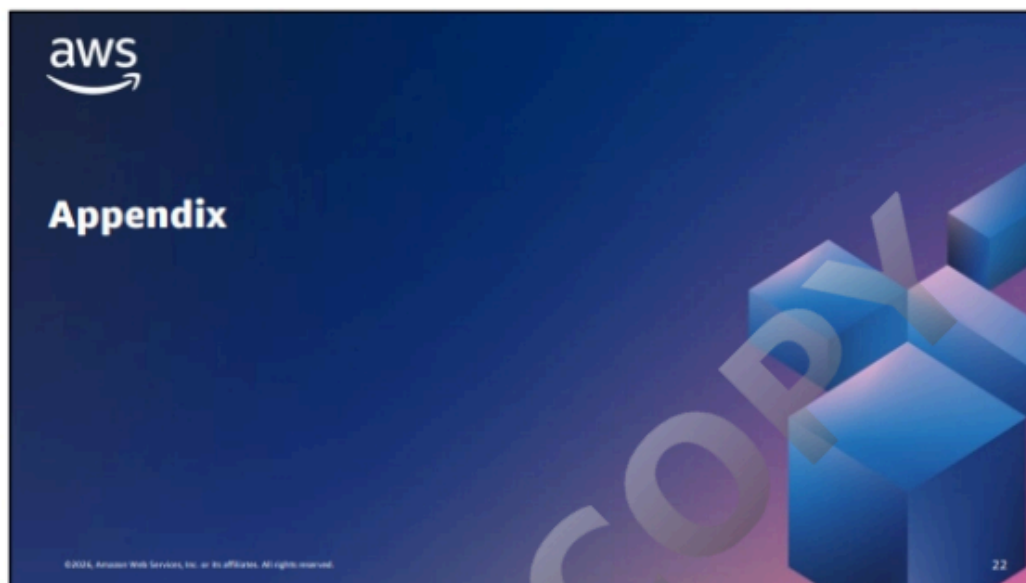
You should now be able to do the following:

- Define generative AI and explain its key capabilities.
- Describe the role of foundation models in generative AI.
- Identify AWS services for implementing generative AI solutions.

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Review the following key takeaways about generative AI and its implementation using AWS services:

- Generative AI represents a paradigm shift in artificial intelligence, with machines creating original content across various mediums. This technology has the potential to transform industries by automating creative tasks, enhancing decision-making processes, and providing more personalized user experiences.
- Foundation models, accessible through services such as Amazon Bedrock, provide a powerful starting point for a wide range of AI applications. You can fine-tune or prompt these models for specific tasks, significantly reducing the time and resources required to develop sophisticated AI solutions.
- AWS offers a comprehensive set of services for you to develop and deploy generative AI. From data preparation and model training with SageMaker AI to accessing pre-trained models through Amazon Bedrock, AWS provides the tools and infrastructure needed to bring generative AI projects from concept to production.
- As you consider implementing generative AI in your own work, remember to balance the exciting possibilities with responsible AI practices. Address challenges such as data privacy, bias mitigation, and ethical considerations to make sure your AI solutions create value while maintaining trust and integrity.
- The field of generative AI is rapidly evolving, with new models and techniques emerging regularly. Staying informed about these developments and continuously experimenting with new approaches will be key to using the full potential of this technology in your business context.



This appendix provides additional resources and references to deepen your understanding of generative AI and AWS services. These materials are designed to support your continued learning and exploration of the topics covered in this module.

References used in this module

- Introducing Generative AI
- Generative AI explained
- AWS generative AI services
- Challenges to consider
- Demonstration
- Module summary



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Module 1: Introducing Generative AI

Introducing Generative AI:

<https://aws.amazon.com/ai/generative-ai/>

AWS generative AI services:

<https://aws.amazon.com/machine-learning/ai-services/>

<https://aws.amazon.com/bedrock/>

<https://aws.amazon.com/sagemaker/>

Challenges to consider:

<https://aws.amazon.com/bedrock/security-compliance/>

https://docs.aws.amazon.com/ja_jp/bedrock/latest/userguide/security.html

Demonstration:

<https://aws.amazon.com/bedrock/>

<https://aws.amazon.com/sagemaker/jumpstart/>

Module summary:

<https://aws.amazon.com/machine-learning/ai-services/>

<https://aws.amazon.com/bedrock/>

<https://aws.amazon.com/sagemaker/>



Welcome to module 2. In this module, you explore how to select business problems for generative AI solutions. This module is designed to equip you with the knowledge and skills to identify when generative AI is the right approach for your organization's challenges.